

EE3621: Digital Signal Processing

Lecture 17: IIR Filter Realization — Direct Forms I, II & Transposed Forms (III B.Tech EEE)

September 9, 2026

1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Formulate** difference equations and signal flow graphs for Direct Form I, Direct Form II, and Transposed Direct Form II.
 2. **Calculate** minimum delay register counts for canonical vs. non-canonical structures.
 3. **Analyze** intermediate node scaling to prevent internal register overflow.
 4. **Compare** Direct Form structures across critical path latency and hardware complexity.
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2. MATHEMATICAL FOUNDATIONS

2.1 General IIR Transfer Function

$$H(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{1 + \sum_{k=1}^N a_k z^{-k}} = \frac{Y(z)}{X(z)}$$

Difference Equation:

$$y[n] = \sum_{k=0}^M b_k x[n-k] - \sum_{k=1}^N a_k y[n-k]$$

2.2 Direct Form I Realization

- Implements feedforward path (M delays) and feedback path (N delays) separately.
- **Total Delays:** $M + N$.
- **Total Multipliers:** $M + N + 1$.
- **Total Adders:** $M + N$.

2.3 Direct Form II (Canonical Realization)

Let $W(z) = \frac{X(z)}{1 + \sum_{k=1}^N a_k z^{-k}}$, then $Y(z) = \left(\sum_{k=0}^M b_k z^{-k} \right) W(z)$. In the time domain:

$$w[n] = x[n] - \sum_{k=1}^N a_k w[n-k]$$

$$y[n] = \sum_{k=0}^M b_k w[n-k]$$

- **Total Delays:** $\max(M, N)$ (Canonical).
- **Total Multipliers:** $M + N + 1$.
- **Total Adders:** $M + N$.

2.4 Transposed Direct Form II

Applying flow graph reversal to Direct Form II:

$$y[n] = b_0 x[n] + v_1[n-1]$$

$$v_k[n] = b_k x[n] - a_k y[n] + v_{k+1}[n-1], \quad k = 1, 2, \dots, N-1$$

$$v_N[n] = b_N x[n] - a_N y[n]$$

Pictorial Representations

IIR Direct Form I realization (Separated Zeros & Poles)



Figure 1: Iir Direct Form I

IIR Direct Form II realization (Shared Delay Line)

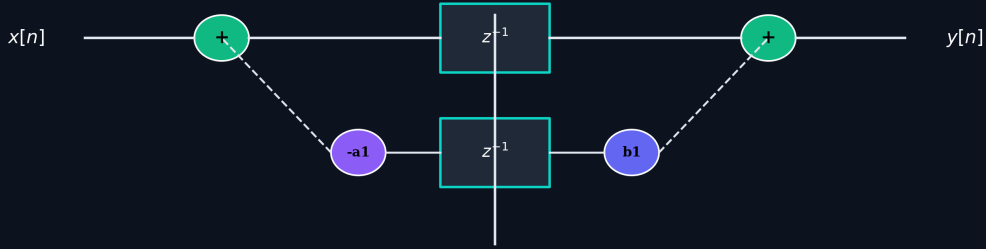


Figure 2: Iir Direct Form Ii

3. WORKED NUMERICAL EXAMPLES

Example 17.1: Direct Form I and II Realization

Problem: Realize the second-order IIR filter:

$$H(z) = \frac{2 + 3z^{-1} + 4z^{-2}}{1 - 0.6z^{-1} + 0.25z^{-2}}$$

in (a) Direct Form I, (b) Direct Form II.

Solution: Here $b_0 = 2, b_1 = 3, b_2 = 4$ and $a_1 = -0.6, a_2 = 0.25$. **(a) Direct Form I:**

- Requires $M + N = 2 + 2 = 4$ delay elements.
- Difference Equation: $y[n] = 2x[n] + 3x[n-1] + 4x[n-2] + 0.6y[n-1] - 0.25y[n-2]$.

(b) Direct Form II (Canonical):

- Requires $\max(2, 2) = 2$ delay elements $w[n-1]$ and $w[n-2]$.
- State Equation: $w[n] = x[n] + 0.6w[n-1] - 0.25w[n-2]$.
- Output Equation: $y[n] = 2w[n] + 3w[n-1] + 4w[n-2]$.

4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

Question 1 (15 Marks)

(a) Realize the system function in Direct Form I and Direct Form II:

$$H(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - 0.75z^{-1} + 0.125z^{-2}}$$

(8 Marks) (b) Derive the Transposed Direct Form II structure for the same transfer function. Why is the transposed structure preferred for high-speed VLSI implementation? *(7 Marks)*

Model Answer & Step-by-Step Marking Rubric:

- **Part (a):**

* Direct Form I SFG (4 delays, 5 multipliers, 4 adders) *(4 Marks)* * Direct Form II Canonical SFG (2 shared delays, state equations) *(4 Marks)*

- **Part (b):**

* Transposed state equations and neatly drawn Transposed DF-II SFG *(4 Marks)* * Explanation: Multipliers share common input $x[n]$ and $y[n]$, adder tree is distributed between delays, eliminating long critical-path propagation delays *(3 Marks)*

5. PYTHON VERIFICATION SCRIPT

```
import scipy.signal as signal

b = [1, 2, 1]
a = [1, -0.75, 0.125]
impulse = [1, 0, 0, 0, 0, 0]
y = signal.lfilter(b, a, impulse)
print("Impulse Response y[n]:", y)
```